Lesson 9.1 - Z-Scores, Confidence Intervals, and Margin of Error

> **Video:** How To Find The Z Score, Confidence Interval, and Margin of Error for a Population Mean by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

Confidence interval for μ (when σ is known)

$$\bar{x} - z\frac{\sigma}{\sqrt{n}} \;<\; \mu \;<\; \bar{x} + z\frac{\sigma}{\sqrt{n}} \qquad \text{or} \qquad \bar{x} \pm \text{EBM}$$

Margin of error, also called the **error bound for the mean (EBM)**:

$$\text{EBM} = z\frac{\sigma}{\sqrt{n}}$$

Finding the critical z-value

The area to the **left** of the positive critical value z is

$$\text{left area} = \frac{\text{CL} + 1}{2}$$

where CL is the confidence level as a decimal. Look this area up in a z-table that gives **left** areas.

Confidence level	Left area	z
90%	0.95	1.645
95%	0.975	1.96

For 90%, the table has 0.94950 at $z = 1.64$ and 0.95053 at $z = 1.65$. 0.95 is about halfway between them, so average: $z = 1.645$.

Example 1 - physics test, 95% CI

$\sigma = 5.4$, $n = 50$, $\bar{x} = 79$

1. z: left area $= (0.95 + 1)/2 = 0.975 \rightarrow z = 1.96$
2. Standard error: $\dfrac{5.4}{\sqrt{50}} \approx 0.76368$
3. **(b) Margin of error:** $\text{EBM} = 1.96 \times 0.76368 \approx 1.4968$
4. **(a) Interval:** 79 ± 1.4968

$$77.5032 < \mu < 80.4968$$

We are 95% confident that the mean score of the whole class lies between about **77.50 and 80.50**.

Example 2 - chemistry test, 90% CI

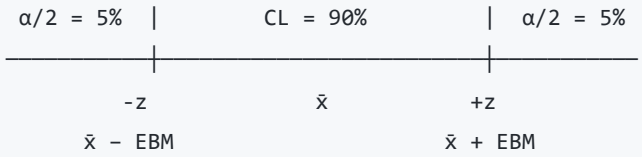
$\sigma = 6.5$, $n = 100$, $\bar{x} = 82$

1. z: left area $= (0.90 + 1)/2 = 0.95 \rightarrow z = 1.645$
2. **(b) Margin of error:** $\text{EBM} = 1.645 \times \dfrac{6.5}{\sqrt{100}} = 1.645 \times 0.65 = 1.06925$
3. **(a) Interval:** 82 ± 1.06925

$$80.93075 < \mu < 83.06925$$

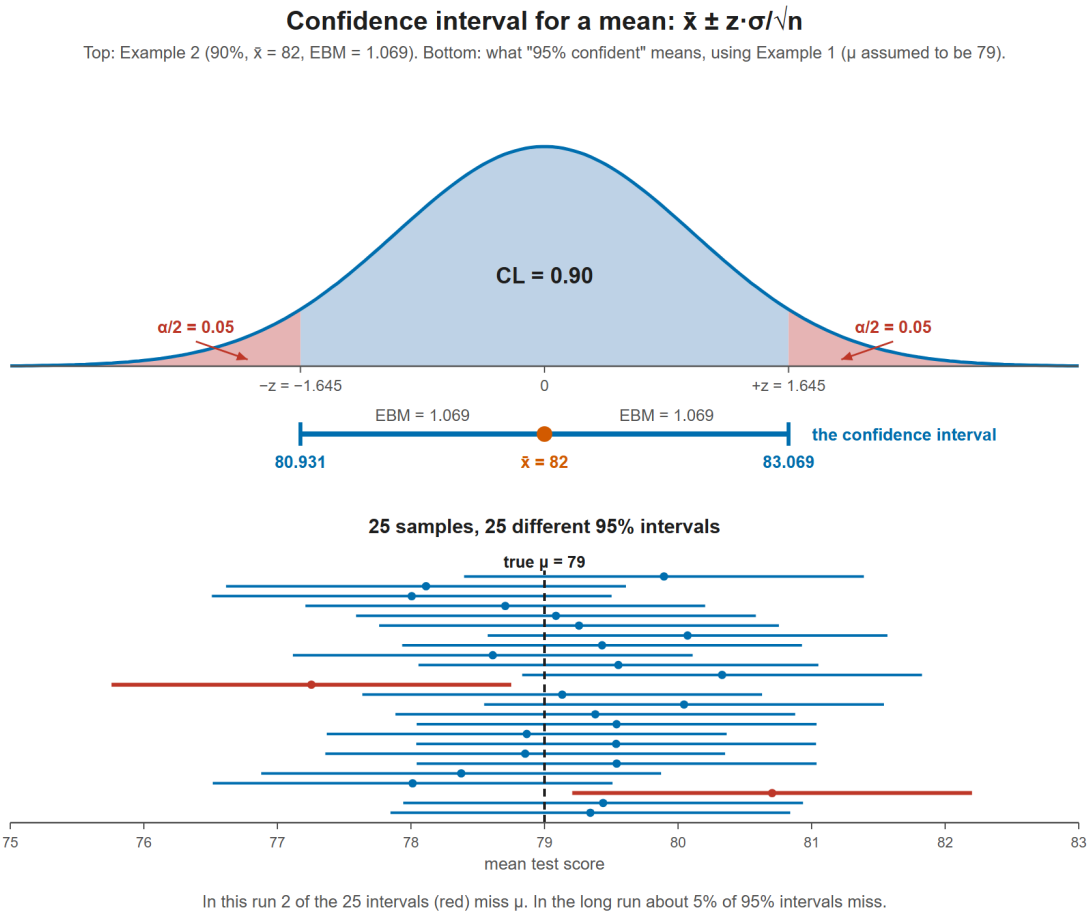
Confidence interval vs confidence level

Term	Meaning	Example
Confidence interval (CI)	The **range of values** $(\bar{x} - \text{EBM},\ \bar{x} + \text{EBM})$	80.93 to 83.07
Confidence level (CL)	The **area under the curve** between $-z$ and $+z$	0.90
α	The area **outside** the interval: $\alpha = 1 - \text{CL}$, split as $\alpha/2$ in each tail	0.10 → 0.05 per tail



Interpreting "95% confidence"

If we took many samples and built a 95% interval from each one, about **95% of those intervals would contain the true mean** μ, and about 5% would miss it. The video phrases this in terms of sample means landing in the interval; this is the more precise statement.



Top: the confidence level is the middle area between $-z$ and $+z$, and the interval reaches one EBM either side of $\bar{x}$. Bottom: 25 simulated samples give 25 different intervals. Most of them cover the true mean (dashed line), but a few miss it, which is what "95% confident" means in the long run.

What changes the margin of error?

$$\text{EBM} = z\,\frac{\sigma}{\sqrt{n}}$$

Change	Effect on EBM	Why
Higher confidence level (90% → 95%)	**Larger**	z grows, so the interval gets wider
Larger sample size n	**Smaller**	$\sqrt{n}$ is in the denominator

To get a smaller margin of error, **lower the confidence level** or **increase the sample size**.

The EBM is half the width of the confidence interval.

Key takeaways

- CI = $\bar{x} \pm z\,\sigma/\sqrt{n}$, and EBM = $z\,\sigma/\sqrt{n}$.
- Critical z: look up the left area $(\text{CL}+1)/2$. 90% → 1.645, 95% → 1.96.
- Higher confidence gives a wider interval. A bigger sample gives a narrower one.

Lesson 9.2 - How to Find the Z-score Given the Confidence Intervals

> **Video:** How To Find The Z Score Given The Confidence Level of a Normal Distribution by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

Common values to memorize

Confidence level	z
90%	1.645
95%	1.96
98%	2.33
99%	2.575

Method for any confidence level

Step 1 - area to the left of $+z$:

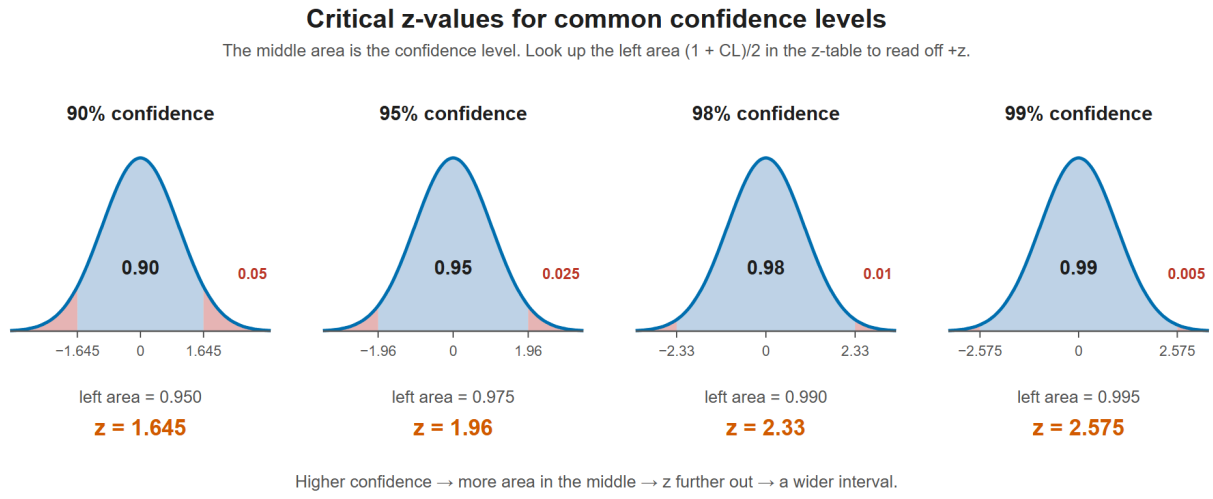
$$\text{left area} = \frac{1 + \text{CL}}{2}$$

Why: the confidence level is the area between $-z$ and $+z$. The leftover $1 - \text{CL}$ is split evenly between the two tails. For 95%:

```
 0.025  |          0.95           |  0.025
--------|-------------------------|--------
       -z           mean         +z
left area up to +z = 0.95 + 0.025 = 0.975
```

Step 2 - find that area inside a positive z-table (one that gives **left** areas). Read the row (ones and tenths digits) and the column (hundredths digit), then add them.

Example: 0.975 sits in row **1.9**, column **0.06**, so $z = 1.9 + 0.06 = 1.96$.



The confidence level is the middle area, and each tail gets half of what is left. The left area up to +z is (1 + CL)/2, which is the number to look up in the z-table.

Practice

97% confidence

$$\text{left area} = \frac{1 + 0.97}{2} = 0.985 \;\Rightarrow\; \text{row } 2.1,\ \text{column } 0.07 \;\Rightarrow\; z = 2.17$$

92% confidence

$$\text{left area} = \frac{1 + 0.92}{2} = 0.96$$

The closest table entry is 0.95994, in row **1.7**, column **0.05**, so $z = 1.75$.

Key takeaways

- Left area $= (1 + \text{CL})/2$. Look it up in the body of the z-table to read off z.
- If the exact area isn't in the table, use the closest entry, or average two entries that are equally close (e.g. 90% → 1.645).
- Know 1.645, 1.96, 2.33 and 2.575 by heart.

Lesson 9.3 - Student's T-Distribution

> **Video:** Student's T Distribution - Confidence Intervals & Margin of Error by The Organic Chemistry Tutor
> **Chapter:** 9 - Confidence Intervals

When to use t instead of z

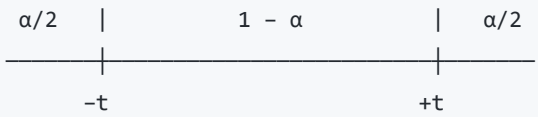
Use the **normal (z)** distribution when...	Use **Student's t** when...
the **population** SD σ is known	only the **sample** SD s is known
$n \geq 30$	$n < 30$

Having only s and a small n are both signs that you need t.

Formulas

$$\bar{x} \pm t_{\nu,\,\alpha/2}\,\frac{s}{\sqrt{n}}, \qquad \text{EBM} = t_{\nu,\,\alpha/2}\,\frac{s}{\sqrt{n}}$$

Symbol	Meaning
$\nu = n - 1$	**Degrees of freedom** (Greek "nu")
$\alpha = 1 - \text{CL}$	The total area outside the interval, i.e. both tails
$\alpha/2 = \dfrac{1 - \text{CL}}{2}$	The **one-tail** area, used to look up t



Reading a t-table: go to the row for ν and the column for the one-tail area $\alpha/2$.

Example 1 - student weights, 95% CI

20 students: $\bar{x} = 165$ lb, $s = 4.5$ lb.

s is a **sample** SD (the school has far more than 20 students) and $n = 20 < 30$, so use t.

- $\nu = 20 - 1 = 19$
- $\alpha/2 = (1 - 0.95)/2 = 0.025$
- Table: $t_{19,\,0.025} = 2.093$

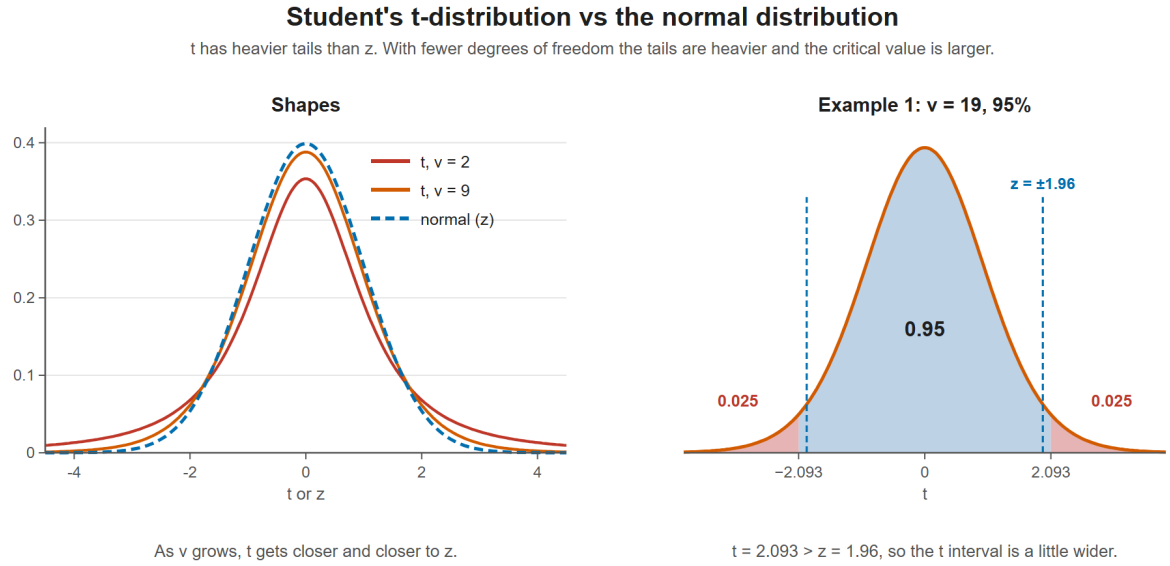
(b) Margin of error

$$\text{EBM} = 2.093 \times \frac{4.5}{\sqrt{20}} = \frac{9.4185}{4.4721} \approx 2.106$$

(a) Interval

$$165 \pm 2.106 \;\Rightarrow\; 162.894 < \mu < 167.106$$

We are 95% confident that the mean weight of all the school's students is between about **162.9 and 167.1 lb**.



Left: t has heavier tails than the normal curve, especially with few degrees of freedom. Right: for $\nu = 19$ the 95% critical value is 2.093, a bit further out than z = 1.96, so t intervals are slightly wider.

Example 2 - chemistry scores, 90% CI

The class has 500 students. 10 scores were sampled: 76, 84, 69, 92, 58, 89, 73, 97, 85, 77.

(a) Sample mean and SD

$$\bar{x} = \frac{800}{10} = 80$$

Distances from 80: 4, 4, 11, 12, 22, 9, 7, 17, 5, 3 (squaring removes the signs).

$$s = \sqrt{\frac{4^2 + 4^2 + 11^2 + 12^2 + 22^2 + 9^2 + 7^2 + 17^2 + 5^2 + 3^2}{10 - 1}} = \sqrt{\frac{1234}{9}} \approx 11.709$$

(b) Margin of error at 90%

- $\nu = 10 - 1 = 9$
- $\alpha/2 = (1 - 0.90)/2 = 0.05$
- Table: $t_{9,\,0.05} = 1.8331$

$$\text{EBM} = 1.8331 \times \frac{11.709}{\sqrt{10}} \approx 6.787$$

(c) 90% confidence interval

$$80 \pm 6.787 \;\Rightarrow\; 73.213 < \mu < 86.787$$

Key takeaways

- Use t when σ is unknown (you only have s), especially when $n < 30$.
- Degrees of freedom $\nu = n - 1$. Look up the one-tail area $\alpha/2 = (1 - \text{CL})/2$.

- CI = $\bar{x} \pm t\, s/\sqrt{n}$. The t value is larger than z for small samples, which gives a wider and more cautious interval.

Lesson 9.4 - Finding a Confidence Interval of a Population Proportion

> **Video:** Finding The Confidence Interval of a Population Proportion Using The Normal Distribution by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

The big idea

You want to know what **fraction** of a large group has some property, for example "what percent of students at this university own a laptop?" Asking everyone (maybe 10,000 people) is too much work, so you ask a **sample** instead.

The sample gives you **one estimate**. Because it's only a sample, that estimate is probably a little off. A confidence interval turns the single estimate into a **range** and says how confident you are that the true value is inside it:

$$\underbrace{\hat{p}}_{\text{your estimate}} \pm \underbrace{E}_{\text{margin of error}}$$

Earlier lessons did this for a **mean** (an average of numbers). This lesson does the same thing for a **proportion** (a percentage of yes answers).

The symbols

The letters are the most confusing part of this lesson, so here they all are in one place:

Symbol	Name	Meaning	Laptop example
n	sample size	how many people you asked	800
x	number of successes	how many said **yes**	584
p	population proportion	the **true** fraction of everyone. Unknown, and it's what we're trying to estimate	?
$\hat{p}$ ("p-hat"), also written p' ("p-prime")	sample proportion	the fraction of **your sample** that said yes: $\hat{p} = x/n$	$584/800 = 0.73$
$\hat{q}$ ("q-hat")	proportion of failures	the fraction of your sample that said **no**: $\hat{q} = 1 - \hat{p}$	0.27
CL	confidence level	how sure you want to be, as a decimal	95% → 0.95
z	critical value	how many standard errors wide the interval is; depends only on CL	1.96
E or EBP	margin of error, or "error bound for a proportion"	how far the interval reaches on each side of $\hat{p}$	0.0308

> **p vs $\hat{p}$:** p (no hat) is the true answer for the whole population, which you never actually know. $\hat{p}$ (with hat) is what your sample gave you. The confidence interval is a range of believable values for p, built around $\hat{p}$.

$\hat{p}$ and $\hat{q}$ always add up to 1. If 73% said yes, then 27% said no.

Formulas

Quantity	Formula
Sample proportion	$\hat{p} = \dfrac{x}{n}$
Proportion of failures	$\hat{q} = 1 - \hat{p}$
Check: is the normal distribution OK?	$n\hat{p}\hat{q} \geq 10$
Area to the left of z	$\dfrac{1 + \mathrm{CL}}{2}$
Margin of error	$E = z\sqrt{\dfrac{\hat{p}\,\hat{q}}{n}}$
Confidence interval	$\hat{p} - E < p < \hat{p} + E$
Sample size needed	$n = \dfrac{z^2\,\hat{p}\,\hat{q}}{E^2}$, always rounded **up**

Common z values:

Confidence level	Area to the left	z
90%	0.95	1.645
95%	0.975	1.96
98%	0.99	2.33
99%	0.995	2.575

The recipe (do these steps in order)

1. **Write down n and x** from the problem.
2. **Find $\hat{p} = x/n$ and $\hat{q} = 1 - \hat{p}$.**
3. **Check $n\hat{p}\hat{q} \geq 10$.** If it is, the normal distribution is a good approximation and you can use z.
4. **Find z** from the confidence level (see the next section).
5. **Find the margin of error $E = z\sqrt{\hat{p}\hat{q}/n}$.**
6. **Build the interval:** lower end $\hat{p} - E$, upper end $\hat{p} + E$.
7. **Write a sentence** saying what it means.

Step 3 in detail - why check $n\hat{p}\hat{q} \geq 10$?

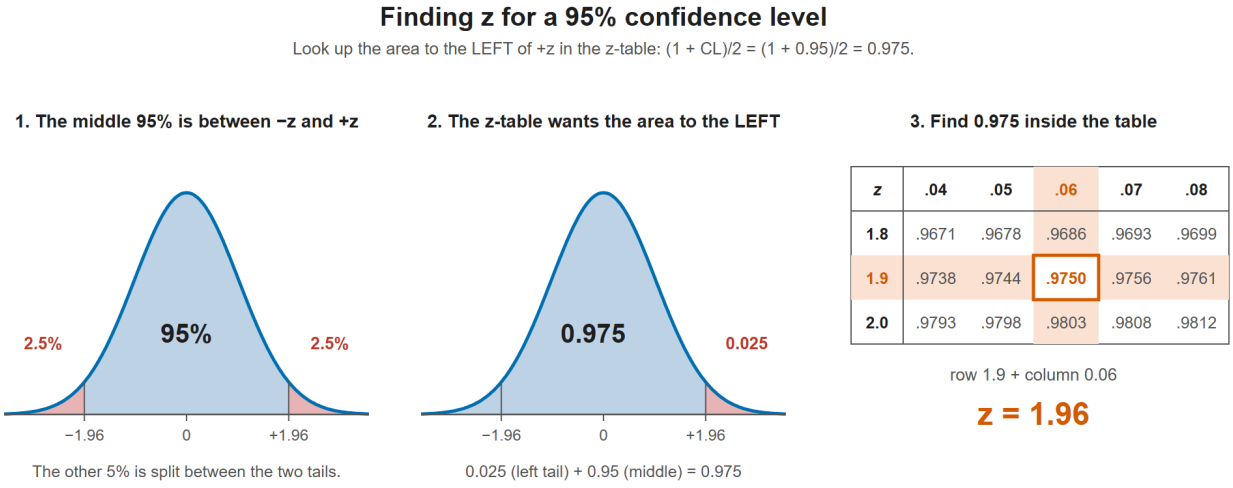
Each person gives one of two answers (yes or no), so the number of yeses follows a **binomial distribution**. The confidence interval formula uses the **normal** (bell) curve instead, because z values come from the normal curve.

That's only allowed when the binomial distribution looks enough like a bell curve, which happens when the sample is big enough and $\hat{p}$ isn't too close to 0 or 1. The rule of thumb $n\hat{p}\hat{q} \geq 10$ tests this.

> Many textbooks use a slightly different check: $n\hat{p} \geq 10$ **and** $n\hat{q} \geq 10$, meaning at least 10 yeses and at least 10 noes. Use whichever your course uses.

Step 4 in detail - finding z from the confidence level

The picture. A 95% confidence level means the middle 95% of the bell curve, between $-z$ and $+z$. The other 5% is split equally between the two tails: 2.5% on the left and 2.5% on the right.



The three steps for 95%. Left: the confidence level is the middle area, and the leftover 5% is split into two 2.5% tails. Middle: the z-table measures from the far left, so the area it wants also includes the left tail, 0.975 in total. Right: 0.9750 sits in row 1.9 and column 0.06, so z = 1.96.

A z-table gives the area to the **left** of a z-score. The area to the left of $+z$ is the middle 95% plus the left tail 2.5%:

$$0.95 + 0.025 = 0.975$$

The shortcut formula gives the same answer without the picture:

$$\text{area to the left} = \frac{1 + \text{CL}}{2} = \frac{1 + 0.95}{2} = \frac{1.95}{2} = 0.975$$

Looking it up in the table. Find 0.975 **inside** the positive z-table, then read off its row and column:

- Row: 1.9
- Column: 0.06
- So $z = 1.9 + 0.06 = 1.96$

If the exact area isn't in the table, take the two closest entries and average their z-scores. For example, for 90% the area to the left is 0.95. The table has 0.9495 (at $z = 1.64$) and 0.9505 (at $z = 1.65$). 0.95 is exactly halfway, so

$$z = \frac{1.64 + 1.65}{2} = 1.645$$

Step 5 in detail - what the margin of error formula means

$$E = z \times \underbrace{\sqrt{\frac{\hat{p}\,\hat{q}}{n}}}_{\text{standard error}}$$

- The square-root part is the **standard error**: roughly how much $\hat{p}$ would vary if you repeated the survey with different random people.
- z sets how many standard errors you allow on each side. More confidence means a bigger z and a wider interval.
- A bigger n makes the standard error smaller, because larger samples are more reliable. That gives a narrower interval.

For why $\hat{p}\hat{q}$ appears here, see "The logic behind the formulas" in Lesson 9.5.

Example 1 - laptop ownership

800 randomly selected college students at university XYZ were asked if they own a laptop. 584 said yes.

(a) Can the binomial distribution be approximated by a normal distribution?

Write down what you know. $n = 800$ and $x = 584$. The true proportion p (all students at the university) is unknown, so we estimate it with $\hat{p}$.

Find $\hat{p}$ and $\hat{q}$.

$$\hat{p} = \frac{x}{n} = \frac{584}{800} = 0.73 \qquad \hat{q} = 1 - 0.73 = 0.27$$

So 73% of the sample said yes and 27% said no.

Check the condition.

$$n\hat{p}\hat{q} = 800(0.73)(0.27) = 157.68 \geq 10 \checkmark$$

Yes, the normal distribution can be used.

(b) 95% confidence interval and margin of error

Find z. The area to the left is $(1 + 0.95)/2 = 0.975$, which gives $z = 1.96$.

Find the margin of error. Work from the inside out:

$$\frac{\hat{p}\,\hat{q}}{n} = \frac{0.73 \times 0.27}{800} = \frac{0.1971}{800} = 2.46375 \times 10^{-4}$$

$$\sqrt{2.46375 \times 10^{-4}} = 0.015696$$

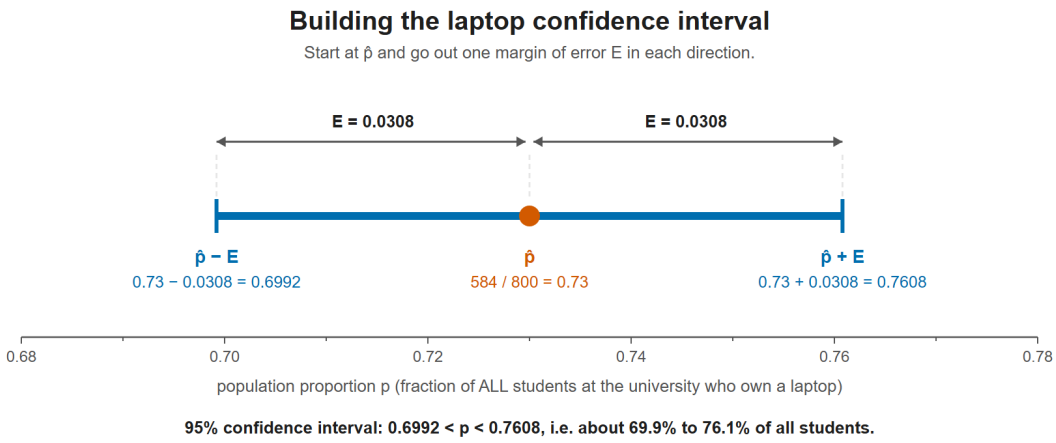
$$E = 1.96 \times 0.015696 = 0.0307648 \approx 0.0308$$

Build the interval.

$$\text{lower} = \hat{p} - E = 0.73 - 0.0308 = 0.6992$$

$$\text{upper} = \hat{p} + E = 0.73 + 0.0308 = 0.7608$$

$$\boxed{0.6992 < p < 0.7608}$$



The interval is built outward from the sample result. The dot is $\hat{p}$ = 0.73, and each arm is one margin of error, so the two ends are $\hat{p} - E = 0.6992$ and $\hat{p} + E = 0.7608$.

Interpretation. We are 95% confident that between about **69.9% and 76.1%** of **all** students at university XYZ own a laptop.

The margin of error is 0.0308, about **±3.1 percentage points**.

(c) What sample size gives a 2% margin of error at 90% confidence?

This time we're **not** building an interval. We're asking how many people we would need to survey.

What changes. $\hat{p} = 0.73$ and $\hat{q} = 0.27$ stay the same. The margin of error is now $E = 2\% = 0.02$, the confidence level is 90%, and n is the unknown.

Solve the margin of error formula for n.

$$E = z\sqrt{\frac{\hat{p}\hat{q}}{n}}$$

Square both sides. The square and the square root cancel:

$$E^2 = \frac{z^2\,\hat{p}\hat{q}}{n}$$

Multiply both sides by n:

$$nE^2 = z^2\,\hat{p}\hat{q}$$

Divide both sides by E^2:

$$n = \frac{z^2\,\hat{p}\,\hat{q}}{E^2}$$

Find z for 90%. The area to the left is $(1 + 0.90)/2 = 0.95$, which gives $z = 1.645$. This is the "average two table values" case from Step 4.

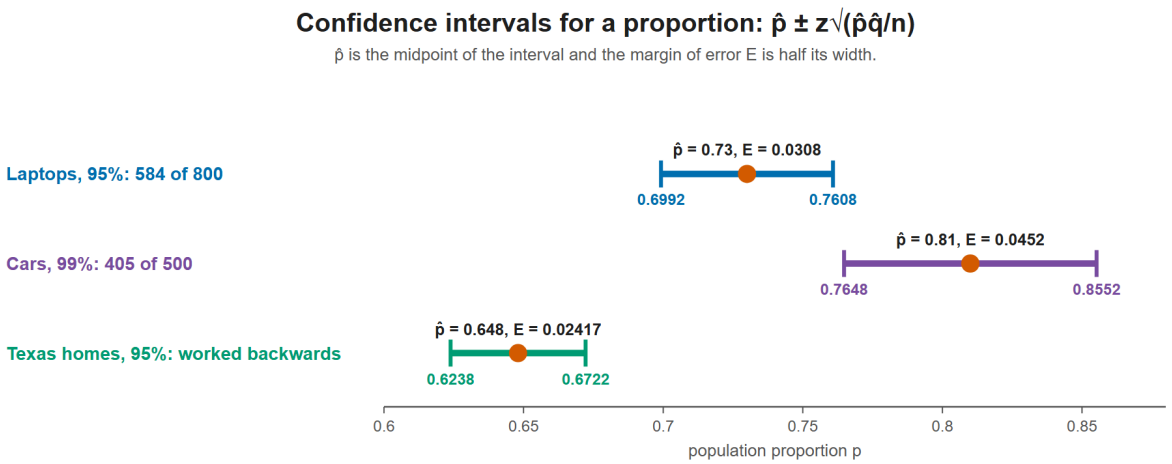
Plug in. Don't forget to square **both** z and E:

$$n = \frac{(1.645)^2(0.73)(0.27)}{(0.02)^2} = \frac{2.706025 \times 0.1971}{0.0004} = \frac{0.53336}{0.0004} \approx 1333.39$$

Round up, even though normal rounding would give 1333:

$$n = 1334 \text{ students}$$

> **Why always round up?** With 1333 students the margin of error would be slightly **more** than 2%. We need **at least** enough people to reach 2%, so we go up to the next whole person.



Texas: p̂ = (0.62383 + 0.67217)/2 = 0.648 and E = (0.67217 − 0.62383)/2 = 0.02417, then n = z²p̂q̂/E² ≈ 1,500.

The three intervals from the lesson. The dot is p̂, the midpoint, and each arm is the margin of error E. The 99% car interval is the widest, because higher confidence uses a larger z.

Example 2 - car ownership

500 residents in town ABC were asked if they own a car. 405 said yes.

Write down what you know and find $\hat{p}$, $\hat{q}$.

$$n = 500, \quad x = 405, \quad \hat{p} = \frac{405}{500} = 0.81, \quad \hat{q} = 1 - 0.81 = 0.19$$

So 81% of the sample own a car.

(a) 99% confidence interval

Find z. The area to the left is $(1 + 0.99)/2 = 1.99/2 = 0.995$. In the table, 0.995 falls between $z = 2.57$ and $z = 2.58$, so average them:

$$z = \frac{2.57 + 2.58}{2} = 2.575$$

Find the margin of error.

$$\frac{0.81 \times 0.19}{500} = \frac{0.1539}{500} = 0.0003078 \qquad \sqrt{0.0003078} = 0.01754$$

$$E = 2.575 \times 0.01754 \approx 0.0452$$

Build the interval.

$$0.81 - 0.0452 = 0.7648 \qquad 0.81 + 0.0452 = 0.8552$$

$$\boxed{0.7648 < p < 0.8552}$$

Interpretation. We are 99% confident that between about **76.5% and 85.5%** of all residents of town ABC own a car.

> Compare with Example 1: the margin of error here (±4.5 points) is wider than the laptop one (±3.1 points). Partly that's because 99% confidence uses a bigger z (2.575 vs 1.96), and partly because the sample is smaller (500 vs 800).

(b) What sample size gives a 3% margin of error at 98% confidence?

Find z. The area to the left is $(1 + 0.98)/2 = 1.98/2 = 0.99$. The closest table entry is 0.9901, at $z = 2.33$.

Plug in, with $E = 3\% = 0.03$:

$$n = \frac{(2.33)^2(0.81)(0.19)}{(0.03)^2} = \frac{5.4289 \times 0.1539}{0.0009} = \frac{0.83551}{0.0009} \approx 928.34$$

Round up: $n = 929$ residents.

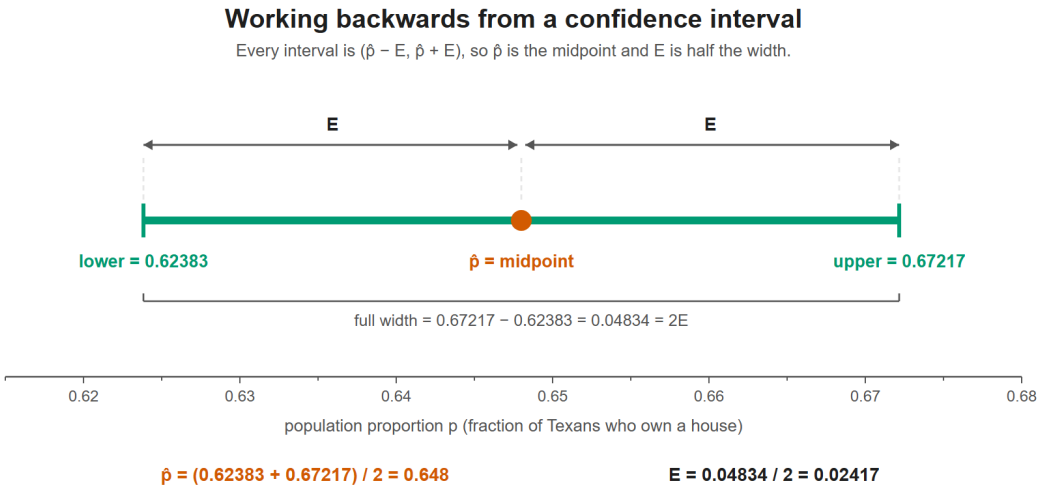
Example 3 - working backwards from an interval

Some number of people in Texas were asked if they own a house. The 95% confidence interval for the true proportion of homeowners was $(0.62383,\ 0.67217)$.

This time the interval is **given**, and we work back to find the pieces that built it. The key fact is that every confidence interval has the shape

$$(\hat{p} - E,\ \ \hat{p} + E)$$

so $\hat{p}$ sits **exactly in the middle**, and E is the distance from the middle to either end.



Only the two ends are given. Because the interval is p̂ ± E, p̂ is the point halfway between them (0.648), and E is half of the full width (0.04834 / 2 = 0.02417).

(a) What is the sample proportion?

$\hat{p}$ is the **midpoint**, the average of the two ends:

$$\hat{p} = \frac{\text{lower} + \text{upper}}{2} = \frac{0.62383 + 0.67217}{2} = \frac{1.296}{2} = 0.648$$

(b) What is the error bound?

Method 1: upper end minus the middle.

$$E = 0.67217 - 0.648 = 0.02417$$

Method 2: half the width of the interval. The full width is 2E, so

$$E = \frac{\text{upper} - \text{lower}}{2} = \frac{0.67217 - 0.62383}{2} = \frac{0.04834}{2} = 0.02417$$

Both methods give the same answer.

(c) How many people were surveyed?

That's n, so use the sample-size formula. At 95% confidence $z = 1.96$, and $\hat{q} = 1 - 0.648 = 0.352$:

$$n = \frac{(1.96)^2(0.648)(0.352)}{(0.02417)^2} = \frac{3.8416 \times 0.228096}{0.000584189} = \frac{0.87625}{0.000584189} \approx 1499.95$$

So about **1500 people** were surveyed.

> Here 1499.95 is just rounding noise in the given interval, so the real answer is 1500. Rounding up and rounding to the nearest whole number both give 1500.

(d) How many said yes?

$\hat{p}$ is the **fraction** who said yes, and x is the **number** who said yes. Since $\hat{p} = x/n$, multiply both sides by n:

$$x = n\,\hat{p} = 1500 \times 0.648 = 972$$

So **972 people** said yes, which is 64.8% of 1500.

Common mistakes

- **Mixing up p and $\hat{p}$.** $\hat{p}$ comes from the sample, and you calculate it. p is the true population value, and you never know it exactly. The interval is a guess for p.
- **Forgetting to convert percentages.** A 2% margin of error is $E = 0.02$, not 2. A 95% confidence level is 0.95.
- **Looking up the confidence level directly in the z-table.** Look up the **area to the left**, $(1 + \text{CL})/2$. For 95%, look up 0.975, not 0.95.
- **Forgetting to square** z or E in the sample-size formula.
- **Rounding n down.** Sample sizes always round **up**.
- **Misreading what "95% confident" means.** It does **not** mean there's a 95% chance that p is in this particular interval. p is a fixed number, and it either is or isn't inside. It means the **method** works 95% of the time: if you repeated the survey many times and built an interval each time, about 95% of those intervals would contain the true p.

Key takeaways

- $\hat{p} = x/n$ is your estimate, and $\hat{q} = 1 - \hat{p}$. Check $n\hat{p}\hat{q} \geq 10$ before using the normal curve.
- Find z by looking up the area to the left, $(1 + \text{CL})/2$, in the z-table.
- Confidence interval: $\hat{p} \pm z\sqrt{\hat{p}\hat{q}/n}$.
- Higher confidence means a bigger z and a wider interval. A bigger sample means a narrower interval.

- Given an interval: its midpoint is $\hat{p}$, and half its width is E.
- Sample size: $n = z^2\hat{p}\hat{q}/E^2$, always rounded **up**. Lesson 9.5 covers sample size in more depth.

Lesson 9.5 - How to Calculate the Sample Size

> **Video:** How To Calculate The Sample Size Given The Confidence Level & Margin of Error by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

Formulas

For a mean. Start from the margin of error:

$$E = z\,\frac{\sigma}{\sqrt{n}} \xrightarrow{\text{square both sides}} E^2 = \frac{z^2\sigma^2}{n} \;\Rightarrow\; \boxed{n = \frac{z^2\sigma^2}{E^2}}$$

For a proportion. Same algebra, starting from the proportion's margin of error (Lesson 9.4):

$$E = z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \xrightarrow{\text{square both sides}} E^2 = \frac{z^2\,\hat{p}(1-\hat{p})}{n} \;\Rightarrow\; \boxed{n = \frac{z^2\,\hat{p}\,(1-\hat{p})}{E^2}}$$

It is the mean formula with σ^2 replaced by $\hat{p}(1-\hat{p})$, as explained below.

If $\hat{p}$ is unknown, use $\hat{p} = 0.5$. That makes $\hat{p}(1-\hat{p})$ as large as possible, so the resulting n is big enough whatever the true proportion is.

Always **round** n **up** to the next whole number.

The logic behind the formulas

1. Every sample gives a different answer. Survey n people and you get a sample proportion $\hat{p}$. A different random group would give a slightly different $\hat{p}$. Across many samples, the $\hat{p}$ values form a bell curve centred on the true p. The spread of that curve is the **standard error**:

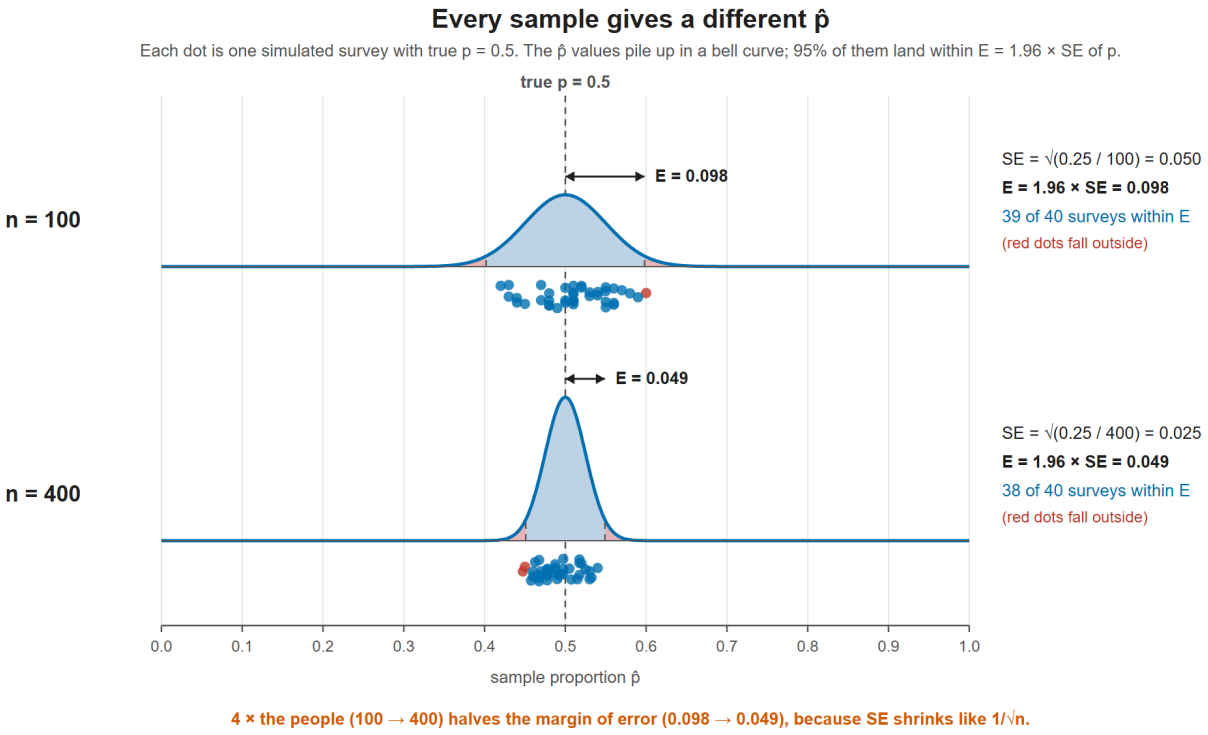
$$SE = \sqrt{\frac{p(1-p)}{n}}$$

Why p here and not $\hat{p}$? The SE describes how much $\hat{p}$ wobbles from sample to sample. That wobble comes from the population the samples are drawn from, so the exact formula uses the true p. But p is exactly what we don't know. In practice we plug in the best stand-in available: the sample's $\hat{p}$ when building an interval from data (Lesson 9.4), or a guess, or 0.5, when planning a survey before any data exists (point 6). That swap is why the formulas at the top are written with $\hat{p}$. With a decent-sized sample $\hat{p}$ is close to p, so the estimated SE is close to the true one.

2. The margin of error is z standard errors. On a bell curve, 95% of values lie within 1.96 standard deviations of the centre. So 95% of the time, $\hat{p}$ lands within $1.96 \times SE$ of the truth:

$$E = z \times SE$$

E is how far off you are willing to be, and z is how sure you want to be of staying within that distance.



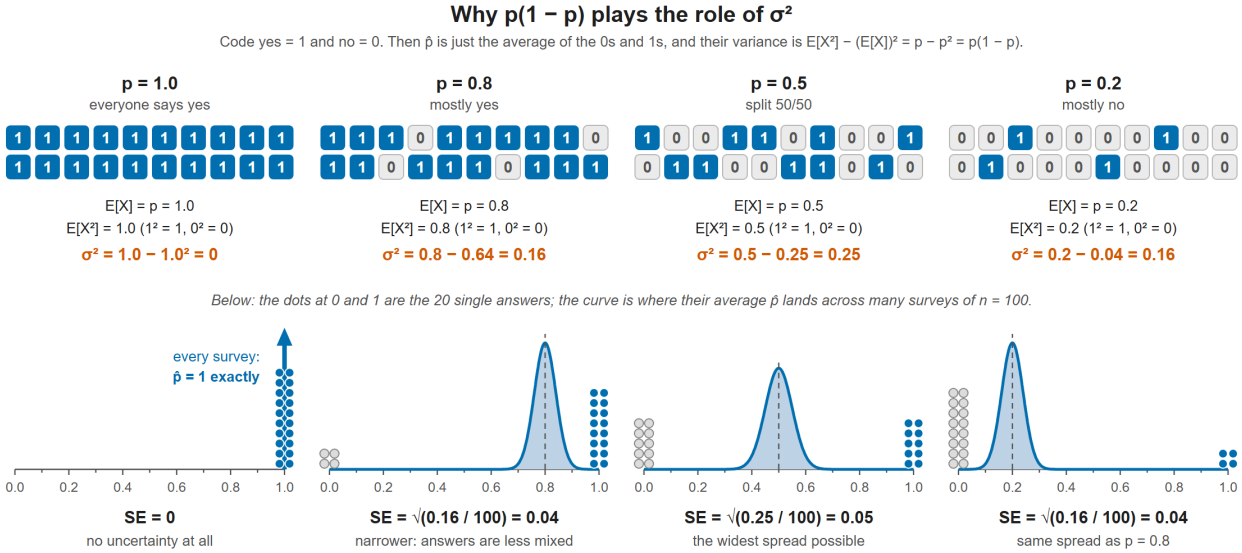
Each dot is one simulated survey. Its $\hat{p}$ misses the true $p = 0.5$ by a different amount each time, but about 95% of the dots land inside $\pm E$. The bottom row previews point 4: four times as many people makes the curve half as wide.

3. Why $p(1-p)$ plays the role of σ^2. Code each answer as 1 (yes) or 0 (no). Then $\hat{p}$ is just the average of those 0s and 1s, and their variance is

$$\sigma^2 = E[X^2] - (E[X])^2 = p - p^2 = p(1-p)$$

since $1^2 = 1$ and $0^2 = 0$. Intuitively:

- If everyone says yes ($p = 1$), every sample looks the same: $p(1-p) = 0$, so there is no uncertainty.
- If the population is split 50/50, samples can easily tilt either way, so the variation is largest: $p(1-p) = 0.25$.

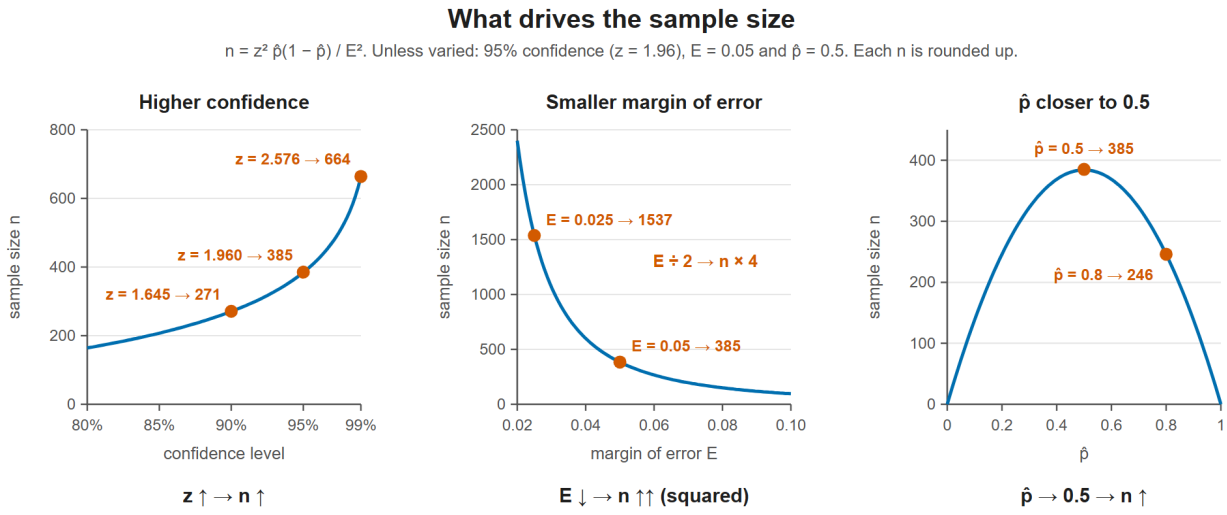


A single answer can only be 0 or 1 (the dots), but the average of 100 answers clusters tightly around p (the curve). The more mixed the 0s and 1s, the bigger $p(1-p)$, and the wider $\hat{p}$ spreads from survey to survey. $p = 0.2$ and $p = 0.8$ give the same spread, since $0.2 \times 0.8 = 0.8 \times 0.2$: only how mixed the answers are matters, not which side is the majority.

4. Why divide by n. In a small sample, a few unusual answers move $\hat{p}$ a lot. In a large one they get diluted. The variance shrinks like $1/n$, so SE shrinks like $1/\sqrt{n}$. Because of the square root, **halving** the margin of error takes **4 times** as many people.

5. Turn it around. Normally you collect data and then compute E. Sample-size planning runs the other way: you choose the E you want and solve for the n that achieves it.

Change	Effect on n	Why
Higher confidence (z ↑)	n ↑	Being more sure needs more evidence
Smaller margin of error (E ↓)	n ↑↑ (squared)	More precision needs more data
p closer to 0.5	n ↑	A more mixed population is less predictable



Each panel changes one ingredient and holds the others at 95%, $E = 0.05$, $\hat{p} = 0.5$. The middle panel is the steepest: halving E multiplies n by 4.

6. Why $\hat{p} = 0.5$ is the safe choice. You plan the survey before you have data, so p is unknown. The safe move is to use the $\hat{p}$ that makes n as large as possible. Since z and E are fixed, that means finding the maximum of

$$f(\hat{p}) = \hat{p}(1-\hat{p}) = \hat{p} - \hat{p}^2, \qquad 0 \le \hat{p} \le 1$$

With calculus. Set the first derivative to zero:

$$f'(\hat{p}) = 1 - 2\hat{p} = 0 \;\Rightarrow\; \hat{p} = 0.5$$

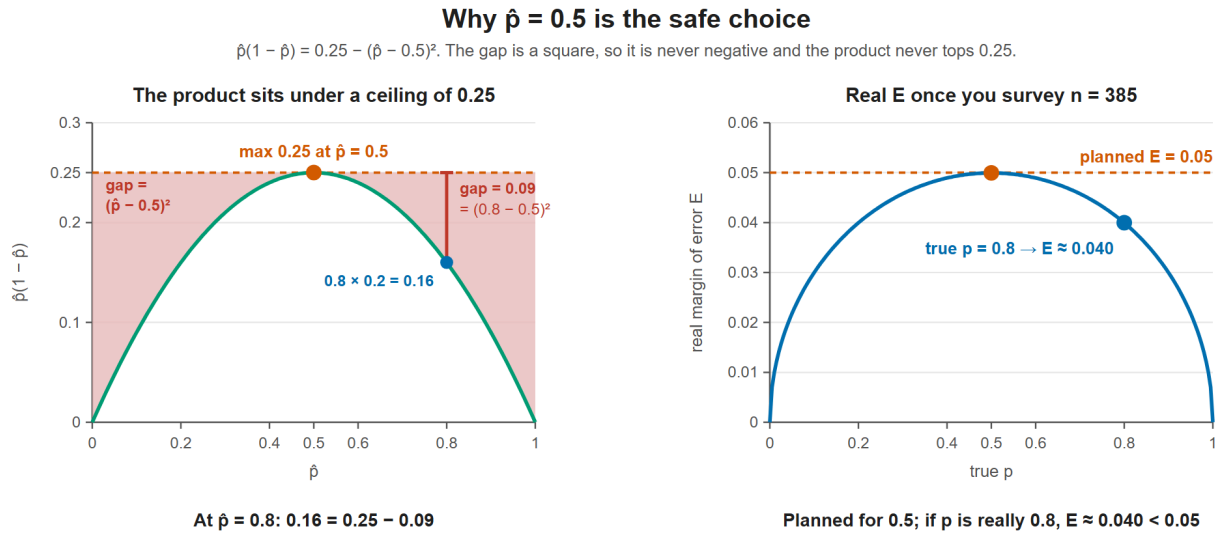
The second derivative is $f''(\hat{p}) = -2 < 0$, so the curve bends downward everywhere and $\hat{p} = 0.5$ is a maximum, not a minimum. Its value is $f(0.5) = 0.5 \times 0.5 = 0.25$. At the ends, $f(0) = f(1) = 0$, so 0.25 is the largest value on the whole range $0 \le \hat{p} \le 1$.

Without calculus. Completing the square gives the same answer:

$$\hat{p}(1-\hat{p}) = 0.25 - (\hat{p} - 0.5)^2$$

The squared term is never negative, so the product can never exceed 0.25, which happens only at $\hat{p} = 0.5$.

Planning for this worst case guarantees n is large enough. If the true p turns out to be, say, 0.8, the real margin of error simply comes out a bit smaller than planned.



Left: the shaded gap between the curve and 0.25 is $(\hat{p} - 0.5)^2$, which is zero only at $\hat{p} = 0.5$. Right: survey the planned $n = 385$, and if p turns out to be 0.8 the margin of error is $1.96\sqrt{0.16/385} \approx 0.040$, comfortably inside the 0.05 you asked for.

> **In one sentence:** sample answers wobble around the truth. The wobble grows with how mixed the population is and shrinks with $\sqrt{n}$, so you pick n large enough that z times the wobble fits inside the error you'll accept.

Example 1 - working backwards to the class size

The mean test score was 84 with standard deviation $\sigma = 15$. The margin of error at 95% confidence was $E = 5.367$. How many students were sampled?

$$n = \frac{(1.96)^2(15)^2}{(5.367)^2} \approx 30.0 \;\Rightarrow\; n = 30 \text{ students}$$

(The mean, 84, isn't needed.)

Example 2 - planning a survey on a proportion

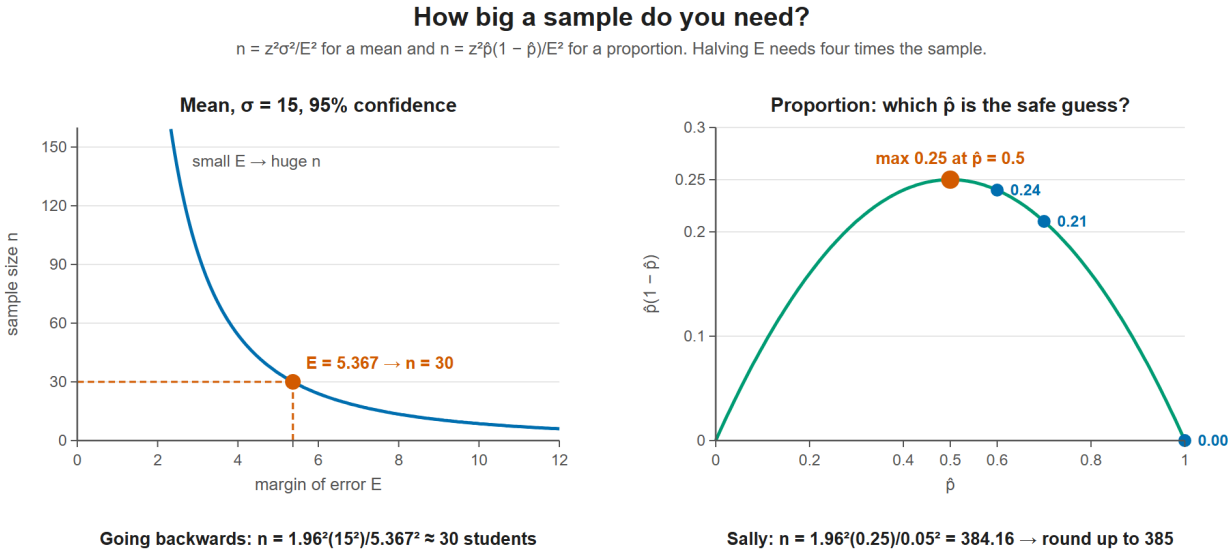
Sally wants to estimate the proportion of high-school students planning to go to college, with **95%** confidence and a **5%** margin of error ($E = 0.05$). She has no prior estimate of $\hat{p}$.

Which $\hat{p}$ makes $\hat{p}(1 - \hat{p})$ largest?

$\hat{p}$	$1-\hat{p}$	Product
1.0	0.0	0.00
0.7	0.3	0.21
0.6	0.4	0.24
0.5	**0.5**	**0.25** (largest)

$$n = \frac{(1.96)^2(0.5)(0.5)}{(0.05)^2} = \frac{0.9604}{0.0025} = 384.16 \;\Rightarrow\; n = 385$$

Round **up**: 384 students would not quite reach the 5% margin.



Left: the required n grows fast as the margin of error shrinks, and Example 1's E = 5.367 corresponds to n = 30. Right: $\hat{p}(1-\hat{p})$ is largest at $\hat{p} = 0.5$, so using 0.5 when nothing is known gives the safest (largest) sample size.

Key takeaways

- Mean: $n = z^2\sigma^2/E^2$. Proportion: $n = z^2\hat{p}(1-\hat{p})/E^2$.
- Both come from solving $E = z \times SE$ for n. For yes/no data the variance is $p(1-p)$, which takes the place of σ^2.
- With no estimate of $\hat{p}$, use 0.5. This gives the largest, safest n.
- Always round **up**.
- A smaller margin of error or a higher confidence level requires a larger sample.

Lesson 9.6 - Chebyshev's Theorem

> **Video:** Chebyshev's Theorem by The Organic Chemistry Tutor **Chapter:** 9 - Confidence Intervals

Statement

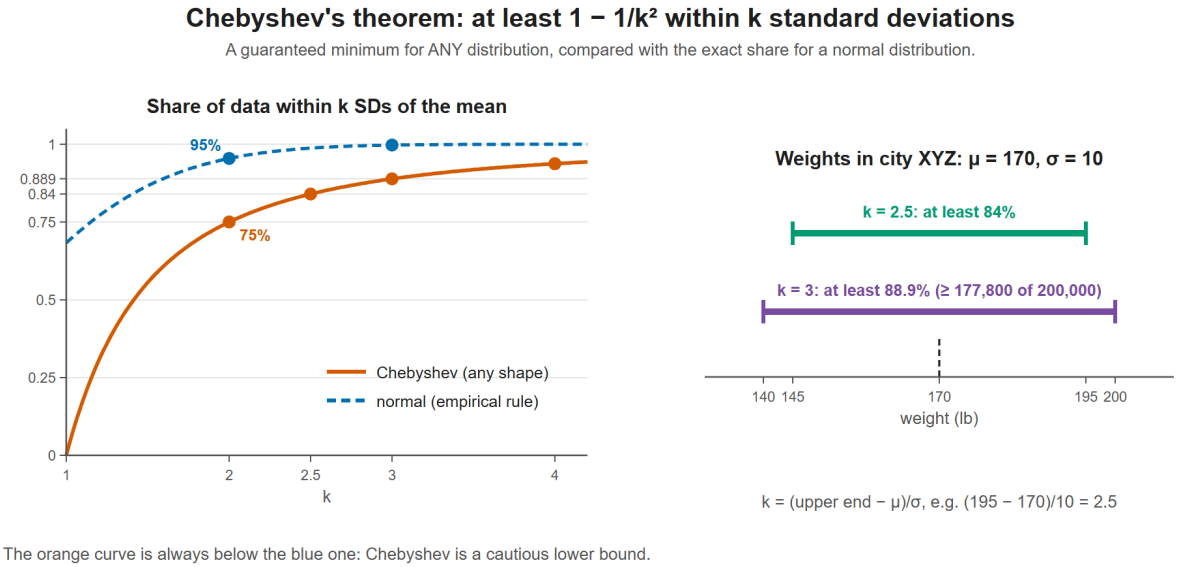
For **any** distribution, whatever its shape, the proportion of data within k standard deviations of the mean is **at least**

$$1 - \frac{1}{k^2}, \qquad k > 1$$

- It gives a **minimum**, not an exact value. The true proportion can be higher.
- It works for **all** distributions, not only normal ones.
- In the video the formula is spoken as "1 over 1 minus k squared". The correct form is $1 - 1/k^2$.

Chebyshev vs the empirical rule

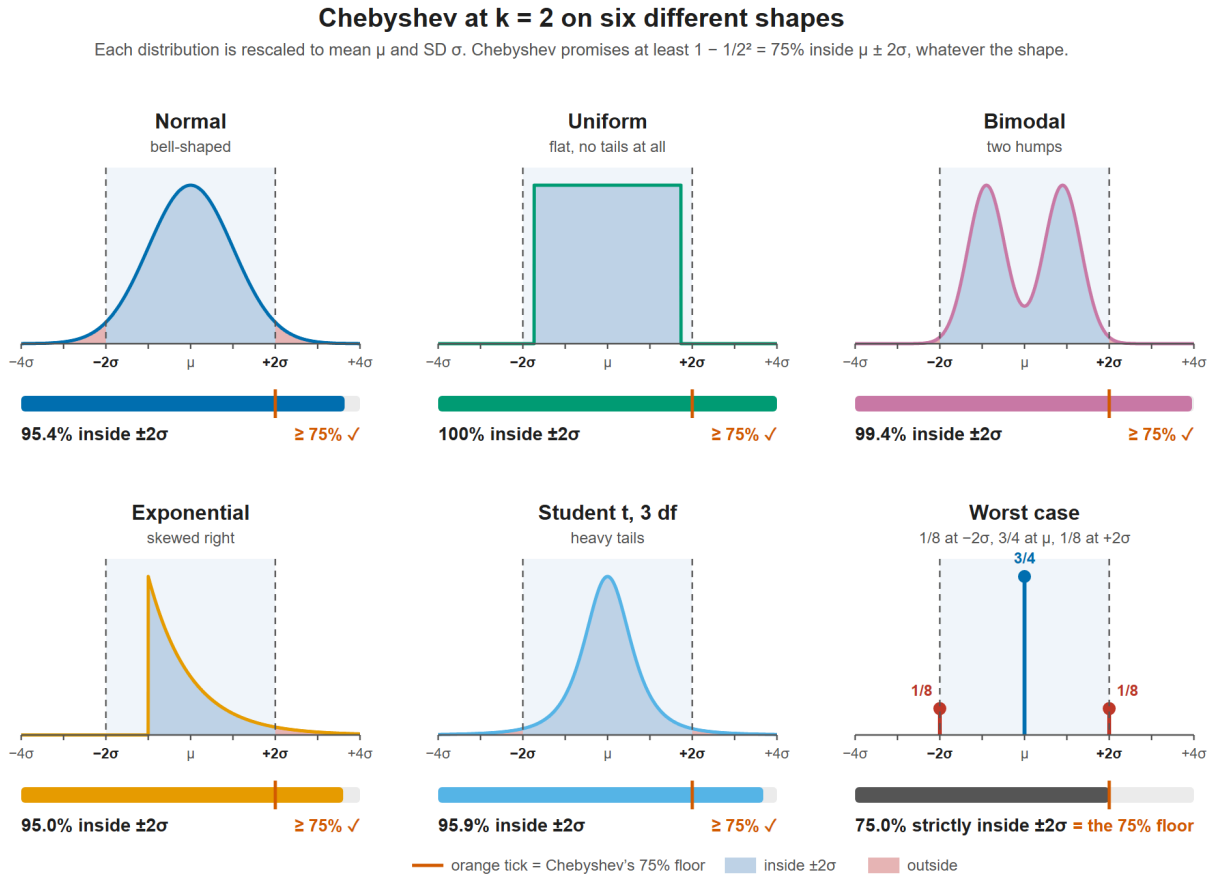
Within	Empirical rule (normal only)	Chebyshev (any distribution, minimum)
1σ	68%	— (k must be > 1)
2σ	95%	$1 - \frac{1}{4} = 75\%$
3σ	99.7%	$1 - \frac{1}{9} \approx 88.9\%$
4σ	≈ 99.99%	$1 - \frac{1}{16} = 93.75\%$



Left: Chebyshev's guaranteed minimum (orange) is always below the exact share for normal data (blue). For example, it guarantees 75% within 2 SDs, while a normal distribution has 95%. Right: the two intervals from the weight example.

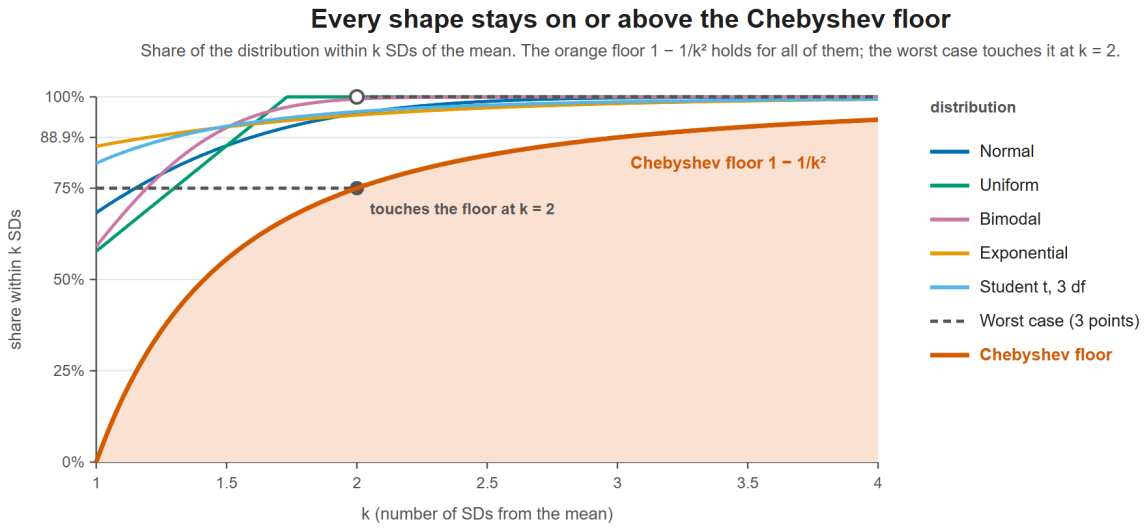
Chebyshev on other famous distributions

"Any distribution" is a strong claim, so here it is tested on six very different shapes. Each one is rescaled to the same mean μ and standard deviation σ, so the axis reads directly in SDs from the mean.



Share within $k\sigma$ of μ	$k = 1.5$	$k = 2$	$k = 3$
Normal	86.6%	95.4%	99.7%
Uniform	86.6%	100%	100%
Bimodal (two humps)	91.6%	99.4%	≈ 100%
Exponential (skewed)	91.8%	95.0%	98.2%
Student t, 3 df (heavy tails)	92.0%	95.9%	98.6%
Worst case (three points)	75%	75%	100%
Chebyshev minimum $1 - 1/k^2$	**55.6%**	**75%**	**88.9%**

- **Every row is at or above the Chebyshev row.** Very different shapes all satisfy the guarantee.
- **Real shapes beat the bound by a lot.** At $k = 2$, five of the six have 95% or more, far above the promised 75%.
- **The bound can't be raised, though.** The worst case puts $\frac{1}{8}$ of the data at $\mu - 2\sigma$, $\frac{3}{4}$ at μ and $\frac{1}{8}$ at $\mu + 2\sigma$. Its SD is exactly σ, since $\sqrt{\frac{1}{8}(2\sigma)^2 + \frac{1}{8}(2\sigma)^2} = \sigma$, yet only 75% lies strictly inside $\mu \pm 2\sigma$. Because one shape lands exactly on 75%, no rule that covers all shapes can promise more.



The orange curve is the floor $1 - 1/k^2$. Each coloured line shows the true share within k SDs for one shape, and none dips below the floor. The dashed worst case touches it at $k = 2$, which is why Chebyshev's theorem is sometimes called "sharp".

What "sharp" means. A bound is *sharp* (or *tight*) when it can't be improved. Suppose someone proposed a stronger rule, such as "at least 80% of any data lies within 2σ". The worst case breaks it, because it has only 75%. So 75% is the best guarantee that holds for every possible shape. Real data usually beats it easily, but no universal rule can promise more.

A worst case for any k. The example above touches the floor only at $k = 2$ because it was built for $k = 2$. The same recipe works for any $k > 1$:

Value	$\mu - k\sigma$	μ	$\mu + k\sigma$
Share of the data	$\frac{1}{2k^2}$	$1 - \frac{1}{k^2}$	$\frac{1}{2k^2}$

- **Its mean is μ**, because the two outer points balance each other.
- **Its SD is exactly σ.** Only the outer points are away from the mean, each by $k\sigma$:

$$\text{variance} = \frac{1}{2k^2}(k\sigma)^2 + \frac{1}{2k^2}(k\sigma)^2 = \frac{\sigma^2}{2} + \frac{\sigma^2}{2} = \sigma^2$$

- **Exactly $1/k^2$ of the data sits $k\sigma$ from the mean**, so the share strictly inside $\mu \pm k\sigma$ is exactly $1 - 1/k^2$, right on the floor.

With $k = 2$ this gives $\frac{1}{8}$, $\frac{3}{4}$, $\frac{1}{8}$, the worst case in the figures. With $k = 3$ it gives $\frac{1}{18}$ at each end and $\frac{8}{9}$ at μ, which touches the floor at 88.9%. So the bound is sharp at every k; each worst-case example just touches the curve at its own k.

Finding k

If an interval is centered on the mean, $\mu \pm k\sigma$, then

$$k = \frac{x - \mu}{\sigma}$$

using the upper end x (the lower end gives the same k).

Example - weights in city XYZ

Mean weight $\mu = 170$ lb, $\sigma = 10$ lb, distribution shape unknown.

(a) Minimum percentage of people weighing between 145 and 195 lb

Both ends are 25 lb from the mean, so

$$k = \frac{195 - 170}{10} = 2.5$$

$$1 - \frac{1}{2.5^2} = 1 - \frac{1}{6.25} = 0.84$$

At least 84% of residents weigh between 145 and 195 lb. The real figure could be 86%, 89%, 95%, ...

(b) The city has 200,000 residents. What is the minimum number weighing between 140 and 200 lb?

Both ends are 30 lb from the mean, so

$$k = \frac{200 - 170}{10} = 3, \qquad 1 - \frac{1}{9} = \frac{8}{9} \approx 0.889$$

$$\frac{8}{9} \times 200{,}000 \approx 177{,}778 \text{ people}$$

(The video rounds to 0.889 first and gets 177,800.)

So $P(140 \le X \le 200) \ge 0.889$, and **at least about 177,800** residents weigh between 140 and 200 lb.

Key takeaways

- Chebyshev: at least $1 - 1/k^2$ of any data set lies within k SDs of the mean ($k > 1$).
- Use it when the distribution is **unknown or not normal**. Use the empirical rule only for normal data.
- $k = (x - \mu)/\sigma$ for an interval centered on the mean.
- Answers are **lower bounds**: say "at least".